

EXPERIMENT 11 CREATING VIEWS

1. Create EMPLOYEE_VU view based on employee numbers, names, and department numbers

```
CREATE OR REPLACE VIEW employee_vu AS
SELECT employee_id, first_name, last_name AS employee,
department_id
FROM employees;
```

Expected Output:

View created.

2. Display the contents of the EMPLOYEE_VU view

```
SELECT * FROM employee_vu;
```

Expected Output:

EMPLOYEE_ID	FIRST_NAME	DEPARTMENT_ID
1	SCOTT	10
2	ADAMS	20
3	JONES	20
4	FORD	20
5	SMITH	20
6	ALLEN	30
7	WARD	30
8	MARTIN	30
9	BLAKE	30
10	CLARK	10
11	TURNER	30
12	POPE	40
13	DESSA	50
14	COOPER	50
15	PERCIVAL	50
16	SMITH	50
17	SMITH	50
18	SMITH	50
19	SMITH	50
20	SMITH	50
21	SMITH	50
22	SMITH	50
23	SMITH	50
24	SMITH	50
25	SMITH	50
26	SMITH	50
27	SMITH	50
28	SMITH	50
29	SMITH	50
30	SMITH	50
31	SMITH	50
32	SMITH	50
33	SMITH	50
34	SMITH	50
35	SMITH	50
36	SMITH	50
37	SMITH	50
38	SMITH	50
39	SMITH	50
40	SMITH	50
41	SMITH	50
42	SMITH	50
43	SMITH	50
44	SMITH	50
45	SMITH	50
46	SMITH	50
47	SMITH	50
48	SMITH	50
49	SMITH	50
50	SMITH	50

```

-----
100 Steven King
90
101 Neena Kochhar
90
102 Lex De Haan
0
103 Alexander Hunold
60
104 Bruce Ernst
60
105 David Austin
60
106 Valli Pataballa
60
107 Diana Lorentz
60
108 Nancy Greenberg
100
109 Daniel Faviet
100
110 John Chen
100
... (continues for all employees)

```

3. Select view name and text from USER_VIEWS

```

SELECT view_name, text
FROM user_views
WHERE view_name = 'EMPLOYEE_VU';

```

Expected Output:

VIEW_NAME	TEXT
EMPLOYEE_VU	SELECT employee_id, first_name, last_name AS employee, department_id FROM employees

4. Using EMPLOYEE_VU view, display all employee names and departments

```

SELECT employee, department_id
FROM employee_vu
ORDER BY department_id;

```

Expected Output:

```

EMPLOYEE DEPARTMENT_ID
-----
Whalen 10 Fay 20 Hartstein 20 Marvis
30 Baida 30 Colmenares 30 Himuro 30
Khoo 30 Raphaely 30 Tobias 30 ...
(continues grouped by department)

```

5. Create DEPT50 view with restrictions

```

CREATE OR REPLACE VIEW dept50 AS
SELECT employee_id AS empno, last_name AS employee,

```

```

department_id AS deptno
FROM employees
WHERE department_id = 50
WITH CHECK OPTION CONSTRAINT dept50_ck;

```

Expected Output:

View created.

6. Display structure and contents of DEPT50 view

```
DESC dept50;
```

Expected Output:

Name	Null?	Type
EMPNO		NUMBER(6)
EMPLOYEE		VARCHAR2(25)
DEPTNO		NUMBER(4)

```

SELECT * FROM dept50;

```

Expected Output:

EMPNO	EMPLOYEE	DEPTNO
120	Weiss	50
121	Fripp	50
122	Kaufling	50
123	Vollman	50
124	Mourgos	50
125	Nayer	50
126	Mikkilineni	50

```
127 Landry 50 128 Markle 50 129 Bissot
50 130 Atkinson 50 ... (all employees
from department 50)
```

7. Attempt to reassign Matos to department 80

```
UPDATE dept50
SET deptno = 80
WHERE employee = 'Matos';
```

Expected Output:

```
ERROR at line 1:
ORA-01402: view WITH CHECK OPTION where-clause violation
```

8. Create SALARY_VU view with employee details, department names, salaries, and grades

```
CREATE OR REPLACE VIEW salary_vu AS
SELECT e.last_name AS "Employee",
       d.department_name AS "Department",
       e.salary AS "Salary",
       jg.grade_level AS "Grade"
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN job_grades jg ON e.salary BETWEEN jg.lowest_sal AND
jg.highest_sal;
```

Expected Output:

View created.

Display SALARY_VU view contents:

```
SELECT * FROM salary_vu ORDER BY "Salary" DESC;
```

Expected Output:

Employee G	Department	Salary

-		
King	Executive	24000
F		
Kochhar	Executive	17000
E		
De Haan	Executive	17000
E		
Hartstein	Marketing	13000
D		
Russell	Sales	14000
D		
Partners	Sales	13500
D		
Errazuriz	Sales	12000
D		
Higgins	Accounting	12000
D		
... (continues with all employees)		

Additional View Examples:

Simple View with Column Aliases

```
CREATE OR REPLACE VIEW emp_simple AS
SELECT employee_id AS id,
       first_name || ' ' || last_name AS full_name,
       salary,
       department_id
FROM employees;
```

Complex View with Aggregation

```
CREATE OR REPLACE VIEW dept_summary AS
SELECT d.department_name,
       COUNT(e.employee_id) AS employee_count,
       AVG(e.salary)        AS avg_salary,
       MAX(e.salary)        AS max_salary,
       MIN(e.salary) AS min_salary
FROM departments d
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_name;
```

Read-Only View

```
CREATE OR REPLACE VIEW emp_readonly AS
SELECT employee_id, first_name, last_name, hire_date
FROM employees
WITH READ ONLY;
```

Key View Concepts Demonstrated:

1. **Simple Views** - Based on single table with no functions
2. **Complex Views** - Contain joins, groups, or functions
3. **Column Aliases** - Renaming columns in views
4. **WITH CHECK OPTION** - Prevents DML that would exclude rows from view

5. **WITH READ ONLY** - Prevents all DML operations on view
6. **Data Dictionary** - USER_VIEWS contains view definitions
7. **View Updating** - Some views allow DML, others don't